$j = 1$ Representation

For $j = 1$ the Lie algebra basis elements are given by :

$$J_x = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad J_y = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{bmatrix}, \quad J_z = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \tag{1}$$

We can write unit vector $\hat{\boldsymbol{n}}$ in terms of polar angles (ψ, ϕ) as follows :

$$\hat{\boldsymbol{n}} = \begin{bmatrix} \sin(\psi)\cos(\phi) \\ \sin(\phi)\sin(\psi) \\ \cos(\psi) \end{bmatrix} \tag{2}$$

Using these forms we now evaluate the following :

$$\begin{aligned}
\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{J}} &= \sin(\psi)\cos(\phi)\, J_x + \sin(\phi)\sin(\psi)\, J_y + \cos(\psi)\, J_z \\
&= \frac{\sin(\psi)\cos(\phi)}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} + \frac{\sin(\phi)\sin(\psi)}{\sqrt{2}} \begin{bmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{bmatrix} + \cos(\psi) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2}\cos(\psi) & e^{-i\phi}\sin(\psi) & 0 \\ e^{i\phi}\sin(\psi) & 0 & e^{-i\phi}\sin(\psi) \\ 0 & e^{i\phi}\sin(\psi) & -\sqrt{2}\cos(\psi) \end{bmatrix}
\end{aligned}$$

We now evaluate :

$$\begin{aligned}
\left(\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{J}}\right)^2 &= \left(\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{J}}\right)\left(\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{J}}\right) \\
&= \frac{1}{2} \begin{bmatrix} \sqrt{2}\cos(\psi) & e^{-i\phi}\sin(\psi) & 0 \\ e^{i\phi}\sin(\psi) & 0 & e^{-i\phi}\sin(\psi) \\ 0 & e^{i\phi}\sin(\psi) & -\sqrt{2}\cos(\psi) \end{bmatrix} \begin{bmatrix} \sqrt{2}\cos(\psi) & e^{-i\phi}\sin(\psi) & 0 \\ e^{i\phi}\sin(\psi) & 0 & e^{-i\phi}\sin(\psi) \\ 0 & e^{i\phi}\sin(\psi) & -\sqrt{2}\cos(\psi) \end{bmatrix} \\
&= \frac{1}{2} \begin{bmatrix} \sin^2(\psi) + 2\cos^2(\psi) & \sqrt{2}e^{-i\phi}\sin(\psi)\cos(\psi) & e^{-2i\phi}\sin^2(\psi) \\ \sqrt{2}e^{i\phi}\sin(\psi)\cos(\psi) & 2\sin^2(\psi) & -\sqrt{2}e^{-i\phi}\sin(\psi)\cos(\psi) \\ e^{2i\phi}\sin^2(\psi) & -\sqrt{2}e^{i\phi}\sin(\psi)\cos(\psi) & \sin^2(\psi) + 2\cos^2(\psi) \end{bmatrix} \\
&= \frac{1}{2} \begin{bmatrix} 1 + \cos^2(\psi) & \sqrt{2}e^{-i\phi}\sin(\psi)\cos(\psi) & e^{-2i\phi}\sin^2(\psi) \\ \sqrt{2}e^{i\phi}\sin(\psi)\cos(\psi) & 2\sin^2(\psi) & -\sqrt{2}e^{-i\phi}\sin(\psi)\cos(\psi) \\ e^{2i\phi}\sin^2(\psi) & -\sqrt{2}e^{i\phi}\sin(\psi)\cos(\psi) & 1 + \cos^2(\psi) \end{bmatrix}
\end{aligned}$$

Where we have used the fact that $\sin^2(\psi) + \cos^2(\psi) = 1$. Once again :

$$R = \left(\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{J}}\right)^3 = \left(\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{J}}\right)\left(\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{J}}\right)^2$$

$$= \frac{1}{2\sqrt{2}} \begin{bmatrix} \sqrt{2}\cos(\psi) & e^{-i\phi}\sin(\psi) & 0 \\ e^{i\phi}\sin(\psi) & 0 & e^{-i\phi}\sin(\psi) \\ 0 & e^{i\phi}\sin(\psi) & -\sqrt{2}\cos(\psi) \end{bmatrix} \times$$

$$\begin{bmatrix} 1+\cos^2(\psi) & \sqrt{2}e^{-i\phi}\sin(\psi)\cos(\psi) & e^{-2i\phi}\sin^2(\psi) \\ \sqrt{2}e^{i\phi}\sin(\psi)\cos(\psi) & 2\sin^2(\psi) & -\sqrt{2}e^{-i\phi}\sin(\psi)\cos(\psi) \\ e^{2i\phi}\sin^2(\psi) & -\sqrt{2}e^{i\phi}\sin(\psi)\cos(\psi) & 1+\cos^2(\psi) \end{bmatrix}$$

$$= \frac{1}{2\sqrt{2}} \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix}$$

Where :

$$\begin{aligned} R_{11} &= \left[\sqrt{2}\cos(\psi)\right] \times \left[1+\cos^2(\psi)\right] + \sqrt{2}\sin^2(\psi)\cos(\psi) \\ &= \sqrt{2}\cos(\psi)\left[1+\cos^2(\psi)+\sin^2(\psi)\right] \\ &= 2\sqrt{2}\cos(\psi) \end{aligned}$$

$$\begin{aligned} R_{12} &= \left[\sqrt{2}\cos(\psi)\right] \times \left[\sqrt{2}e^{-i\phi}\sin(\psi)\cos(\psi)\right] + 2\,e^{-i\phi}\sin^3(\psi) \\ &= 2\,e^{-i\phi}\sin(\psi)\cos^2(\psi) + 2\,e^{-i\phi}\sin^3(\psi) \\ &= 2\,e^{-i\phi}\sin(\psi)\left[\cos^2(\psi)+\sin^2(\psi)\right] \\ &= 2\,e^{-i\phi}\sin(\psi) \end{aligned}$$

$$R_{13} = \sqrt{2}\,e^{-2i\phi}\cos(\psi)\sin^2(\psi) - \sqrt{2}e^{-2i\phi}\sin^2(\psi)\cos(\psi) = 0$$

$$\begin{aligned} R_{21} &= \left[e^{i\phi}\sin(\psi)\right] \times \left[1+\cos^2(\psi)\right] + \left[e^{-i\phi}\sin(\psi)\right] \times \left[e^{2i\phi}\sin^2(\psi)\right] \\ &= e^{i\phi}\sin(\psi)\left[1+\cos^2(\psi)+\sin^2(\psi)\right] \\ &= 2e^{i\phi}\sin(\psi) \end{aligned}$$

$$\begin{aligned} R_{22} &= \left[e^{i\phi}\sin(\psi)\right] \times \left[\sqrt{2}e^{-i\phi}\sin(\psi)\cos(\psi)\right] + \left[e^{-i\phi}\sin(\psi)\right] \times \left[-\sqrt{2}e^{i\phi}\sin(\psi)\cos(\psi)\right] \\ &= \sqrt{2}\sin^2(\psi)\cos(\psi) - \sqrt{2}\sin^2(\psi)\cos(\psi) = 0 \end{aligned}$$

$$R_{23} = \left[e^{i\phi}\sin(\psi)\right] \times \left[e^{-2i\phi}\sin^2(\psi)\right] + \left[e^{-i\phi}\sin(\psi)\right] \times \left[1+\cos^2(\psi)\right]$$

$$= e^{-i\phi} \sin^3(\psi) + e^{-i\phi} \sin(\psi) \left[1 + \cos^2(\psi)\right]$$

$$= e^{-i\phi} \sin(\psi) \left[\sin^2(\psi) + 1 + \cos^2(\psi)\right]$$

$$= 2e^{-i\phi} \sin(\psi)$$

$$R_{31} = \left[e^{i\phi} \sin(\psi)\right] \times \left[\sqrt{2}e^{i\phi} \sin(\psi) \cos(\psi)\right] + \left[-\sqrt{2}\cos(\psi)\right] \times \left[e^{2i\phi} \sin^2(\psi)\right]$$

$$= \sqrt{2}\, e^{2i\phi} \sin^2(\psi) \cos(\psi) - \sqrt{2}\, e^{2i\phi} \cos(\psi) \sin^2(\psi) = 0$$

$$R_{32} = \left[e^{i\phi} \sin(\psi)\right] \times \left[2\sin^2(\psi)\right] + \left[-\sqrt{2}\cos(\psi)\right] \times \left[-\sqrt{2}e^{i\phi} \sin(\psi) \cos(\psi)\right]$$

$$= 2\, e^{i\phi} \sin^3(\psi) + 2\, e^{i\phi} \sin(\psi) \cos^2(\psi)$$

$$= 2\, e^{i\phi} \sin(\psi) \left[\sin^2(\psi) + \cos^2(\psi)\right]$$

$$= 2\, e^{i\phi} \sin(\psi)$$

$$R_{33} = \left[e^{i\phi} \sin(\psi)\right] \times \left[-\sqrt{2}e^{-i\phi} \sin(\psi) \cos(\psi)\right] + \left[-\sqrt{2}\cos(\psi)\right] \times \left[1 + \cos^2(\psi)\right]$$

$$= -\sqrt{2}\, \sin^2(\psi) \cos(\psi) - \sqrt{2}\cos(\psi) \times \left[1 + \cos^2(\psi)\right]$$

$$= -\sqrt{2}\, \cos(\psi) \left[\sin^2(\psi) + 1 + \cos^2(\psi)\right]$$

$$= -2\sqrt{2}\, \cos(\psi)$$

Inserting which gives us :

$$\begin{aligned}
R &= \left(\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{J}}\right)^3 \\
&= \frac{1}{2\sqrt{2}} \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix} \\
&= \frac{1}{2\sqrt{2}} \begin{bmatrix} 2\sqrt{2}\cos(\psi) & 2\, e^{-i\phi} \sin(\psi) & 0 \\ 2e^{i\phi} \sin(\psi) & 0 & 2e^{-i\phi} \sin(\psi) \\ 0 & 2\, e^{i\phi} \sin(\psi) & -2\sqrt{2}\, \cos(\psi) \end{bmatrix} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2}\cos(\psi) & e^{-i\phi} \sin(\psi) & 0 \\ e^{i\phi} \sin(\psi) & 0 & e^{-i\phi} \sin(\psi) \\ 0 & e^{i\phi} \sin(\psi) & -\sqrt{2}\cos(\psi) \end{bmatrix} = \left(\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{J}}\right)
\end{aligned}$$

We now have the following identity :

$$\boxed{\left(\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{J}}\right)^3 = \left(\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{J}}\right)} \tag{3}$$

Which gives us the following relation :

$$\left(\hat{\boldsymbol{n}}\cdot\vec{\boldsymbol{J}}\right)^{2m+1} = \left(\hat{\boldsymbol{n}}\cdot\vec{\boldsymbol{J}}\right) \tag{4}$$

$$\left(\hat{\boldsymbol{n}}\cdot\vec{\boldsymbol{J}}\right)^{2m} = \left(\hat{\boldsymbol{n}}\cdot\vec{\boldsymbol{J}}\right)^{2} \tag{5}$$

Where $m \in \mathbb{Z}_{>0}$, i.e. $m = \{1, 2, 3, \dots\}$.

We now are in a position to write the representation of any element in SU(2) corresponding to $j = 1$ representation., which is given by :

$$\rho(g) = \mathbf{exp}\left[-i\theta\left(\hat{\boldsymbol{n}}\cdot\vec{\boldsymbol{J}}\right)\right] \tag{6}$$

Expanding Eq. 6 gives :

$$\begin{aligned}
\rho(g) &= \mathbb{1}_{3\times3} - i\theta\left(\hat{\boldsymbol{n}}\cdot\vec{\boldsymbol{J}}\right) + \frac{\left[i\theta\left(\hat{\boldsymbol{n}}\cdot\vec{\boldsymbol{J}}\right)\right]^2}{2!} - \frac{\left[i\theta\left(\hat{\boldsymbol{n}}\cdot\vec{\boldsymbol{J}}\right)\right]^3}{3!} + \cdots \\
&= \mathbb{1}_{3\times3} + \sum_{n=1}^{\infty}\frac{(-i\theta)^n}{n!}\left(\hat{\boldsymbol{n}}\cdot\vec{\boldsymbol{J}}\right)^n \\
&= \mathbb{1}_{3\times3} + \left[\sum_{p=0}^{\infty}\frac{(-i\theta)^{2p+1}}{(2p+1)!}\left(\hat{\boldsymbol{n}}\cdot\vec{\boldsymbol{J}}\right)^{(2p+1)}\right] + \left[\sum_{q=1}^{\infty}\frac{(-i\theta)^{2q}}{(2q)!}\left(\hat{\boldsymbol{n}}\cdot\vec{\boldsymbol{J}}\right)^{2q}\right] \\
&= \mathbb{1}_{3\times3} + \left(\hat{\boldsymbol{n}}\cdot\vec{\boldsymbol{J}}\right)\left[\sum_{p=0}^{\infty}\frac{(-i\theta)^{2p+1}}{(2p+1)!}\right] + \left(\hat{\boldsymbol{n}}\cdot\vec{\boldsymbol{J}}\right)^2\left[\sum_{q=1}^{\infty}\frac{(-i\theta)^{2q}}{(2q)!}\right]
\end{aligned}$$

We now evaluate these two sums one by one. First :

$$\begin{aligned}
\sum_{p=0}^{\infty}\frac{(-i\theta)^{2p+1}}{(2p+1)!} &= -i\theta + \sum_{p=1}^{\infty}\frac{(-i\theta)^{2p+1}}{(2p+1)!} \\
&= -i\theta + \sum_{p=1,3,5,\dots}^{\infty}\frac{(-i\theta)^{2p+1}}{(2p+1)!} + \sum_{p=2,4,6,\dots}^{\infty}\frac{(-i\theta)^{2p+1}}{(2p+1)!} \\
&= -i\theta + i\left(\sum_{p=1,3,5,\dots}^{\infty}\frac{(\theta)^{2p+1}}{(2p+1)!}\right) - i\left(\sum_{p=2,4,6,\dots}^{\infty}\frac{(\theta)^{2p+1}}{(2p+1)!}\right) \\
&= -i\theta + i\left[\frac{\theta^3}{3!} + \frac{\theta^7}{7!} + \cdots\right] - i\left[\frac{\theta^5}{5!} + \frac{\theta^9}{9!} + \cdots\right] \\
&= -i\left[\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \frac{\theta^9}{9!} - \cdots\right] \\
&= -i\sin\theta
\end{aligned}$$

Where we have used the following arguments : $(-i)^{2m} = (-1)^{2m}\cdot(i)^{2m} = i^{2m} = -1$ if m is odd, and 1 if m is even. Shifting our result, we can say $(-i)^{2r+1} = (-i)^{2r}\cdot(-i) = i$ if r is odd, and $-i$ if r is even.

The next sum is evaluated as follows :

$$
\begin{aligned}
\sum_{q=1}^{\infty} \frac{(-i\theta)^{2q}}{(2q)!} &= \sum_{q=1,3,5,\dots}^{\infty} \frac{(-i\theta)^{2q}}{(2q)!} + \sum_{q=2,4,6,\dots}^{\infty} \frac{(-i\theta)^{2q}}{(2q)!} \\
&= -1 \left(\sum_{q=1,3,5,\dots}^{\infty} \frac{(\theta)^{2q}}{(2q)!} \right) + 1 \left(\sum_{q=2,4,6,\dots}^{\infty} \frac{(\theta)^{2q}}{(2q)!} \right) \\
&= -1 \left[\frac{\theta^2}{2!} + \frac{\theta^6}{6!} + \cdots \right] + 1 \left[\frac{\theta^4}{4!} + \frac{\theta^8}{8!} + \cdots \right] \\
&= -\frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \frac{\theta^8}{8!} - \cdots \\
&= 1 - 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \frac{\theta^8}{8!} - \cdots \\
&= -1 + \left[1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \frac{\theta^8}{8!} - \cdots \right] \\
&= -1 + \cos\theta
\end{aligned}
$$

Which gives us the representation as :

$$
\boxed{\rho(g) = \mathbb{1}_{3\times 3} - i \left(\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{J}} \right) \sin\theta + (\cos\theta - 1) \left(\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{J}} \right)^2} \tag{7}
$$

Which is quite nice form. Substituting the form of $\left(\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{J}} \right)$ and $\left(\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{J}} \right)^2$ in this we get :

$$
\rho(g) = \mathbb{1}_{3\times 3} - \frac{i \sin\theta}{\sqrt{2}} \begin{bmatrix} \sqrt{2}\cos(\psi) & e^{-i\phi}\sin(\psi) & 0 \\ e^{i\phi}\sin(\psi) & 0 & e^{-i\phi}\sin(\psi) \\ 0 & e^{i\phi}\sin(\psi) & -\sqrt{2}\cos(\psi) \end{bmatrix} +
$$

$$
\frac{(\cos\theta - 1)}{2} \begin{bmatrix} 1 + \cos^2(\psi) & \sqrt{2}e^{-i\phi}\sin(\psi)\cos(\psi) & e^{-2i\phi}\sin^2(\psi) \\ \sqrt{2}e^{i\phi}\sin(\psi)\cos(\psi) & 2\sin^2(\psi) & -\sqrt{2}e^{-i\phi}\sin(\psi)\cos(\psi) \\ e^{2i\phi}\sin^2(\psi) & -\sqrt{2}e^{i\phi}\sin(\psi)\cos(\psi) & 1 + \cos^2(\psi) \end{bmatrix}
$$

Which gives us :

$$
\rho(g) = \begin{bmatrix} \frac{(\cos^2(\psi)+1)(\cos(\theta)-1)}{2} - i\sin(\theta)\cos(\psi) + 1 & \frac{\sqrt{2}((\cos(\theta)-1)\cos(\psi)-i\sin(\theta))e^{-i\phi}\sin(\psi)}{2} & \frac{(\cos(\theta)-1)e^{-2i\phi}\sin^2(\psi)}{2} \\ \frac{\sqrt{2}((\cos(\theta)-1)\cos(\psi)-i\sin(\theta))e^{i\phi}\sin(\psi)}{2} & (\cos(\theta)-1)\sin^2(\psi) + 1 & \frac{\sqrt{2}(-(\cos(\theta)-1)\cos(\psi)-i\sin(\theta))e^{-i\phi}\sin(\psi)}{2} \\ \frac{(\cos(\theta)-1)e^{2i\phi}\sin^2(\psi)}{2} & \frac{\sqrt{2}(-(\cos(\theta)-1)\cos(\psi)-i\sin(\theta))e^{i\phi}\sin(\psi)}{2} & \frac{(\cos^2(\psi)+1)(\cos(\theta)-1)}{2} + i\sin(\theta)\cos(\psi) + 1 \end{bmatrix}
$$

$$\rho(g) = \begin{bmatrix} \frac{(\cos^2(\psi)+1)(\cos(\theta)-1)}{2} - i\sin(\theta)\cos(\psi) + 1 & \frac{\sqrt{2}((\cos(\theta)-1)\cos(\psi)-i\sin(\theta))e^{-i\phi}\sin(\psi)}{2} & \frac{(\cos(\theta)-1)e^{-2i\phi}\sin^2(\psi)}{2} \\ \frac{\sqrt{2}((\cos(\theta)-1)\cos(\psi)-i\sin(\theta))e^{i\phi}\sin(\psi)}{2} & (\cos(\theta)-1)\sin^2(\psi)+1 & \frac{\sqrt{2}(-(\cos(\theta)-1)\cos(\psi)-i\sin(\theta))e^{-i\phi}\sin(\psi)}{2} \\ \frac{(\cos(\theta)-1)e^{2i\phi}\sin^2(\psi)}{2} & \frac{\sqrt{2}(-(\cos(\theta)-1)\cos(\psi)-i\sin(\theta))e^{i\phi}\sin(\psi)}{2} & \frac{(\cos^2(\psi)+1)(\cos(\theta)-1)}{2} + i\sin(\theta)\cos(\psi) + 1 \end{bmatrix} \tag{8}$$